

THE SIALIC-GLYCAN INFLAMMATORY CASCADE (SGIC):

A Unified Framework for the Role of Dietary Neu5Gc and Wheat-Derived Xenoantigens in Chronic Inflammatory Disease

Master Overview — Hamer Glycan Systems White Paper Series

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Preamble: How to Read This Document

This document is the master overview of the Hamer Glycan Systems research framework. It does not replace any of the individual white papers or the Hamer Loop Manuscript Series — it maps the territory they collectively cover and explains how they connect to each other.

The framework has a simple spine: two specific dietary antigens — N-glycolylneuraminic acid (Neu5Gc) from bovine red meat and dairy, and wheat-derived amylase-trypsin inhibitors (ATIs) and gliadins — when consumed chronically in the quantities characteristic of a Western dietary pattern, produce a cascade of immune and inflammatory consequences that we term the Sialic-Glycan Inflammatory Cascade (SGIC). This cascade has a core mechanism and a set of downstream branches. The core mechanism is the same in every individual susceptible to it. The downstream branch that becomes most clinically visible depends on the individual's genetic background, immune phenotype, and the specific pattern and severity of their dietary antigen exposure.

The skin is the most visible branch. But it is not the only one, and in many individuals it is not the most consequential one. This document maps all of the branches currently identified within the framework, explains the shared upstream mechanism that connects them, and provides a guide to the individual papers that develop each branch in full mechanistic detail.

The framework is built on 45 years of longitudinal self-observation, pattern recognition, and systematic provocation-elimination trials by the author, supplemented by the established published literature in glycobiology, gastroenterology, immunology, and the relevant clinical specialties. It is offered as a hypothesis framework for scientific investigation, not as established clinical fact. Every claim is hedged accordingly in the individual papers. This overview does not repeat those hedges in full but asks the reader to hold them in mind throughout.

1. The Two Antigens

1.1 Neu5Gc: The Evolutionary Misfit

N-glycolylneuraminic acid (Neu5Gc) is a sialic acid produced by virtually all mammals through the enzyme CMP-Neu5Ac hydroxylase (CMAH). Approximately 2–3 million years ago, a mutation inactivated the CMAH gene in the human lineage. Every human alive today lacks the ability to synthesise Neu5Gc. We are the only great ape for whom this is true.

The reason for this mutation is believed to be pathogen-driven — an ancestral malaria-like organism used Neu5Gc as a binding site on red blood cells. The mutation that removed Neu5Gc from the human surface conferred survival advantage. But the tradeoff was permanent: a species that cannot produce Neu5Gc, has no immune tolerance for it, and yet consumes it in large quantities every time it eats red meat or bovine dairy.

When dietary Neu5Gc is absorbed from food, it is incorporated into human cell surface glycoproteins. The immune system recognises it as foreign — because it is foreign — and produces anti-Neu5Gc antibodies that engage the incorporated antigen at tissue surfaces throughout the body. This creates a state of chronic, low-grade, multi-tissue inflammation called xenosialitis. It never fully resolves while Neu5Gc is being consumed. It activates NF- κ B, elevates TNF- α , IL-6, and IL-1 β , damages intestinal tight junctions, impairs vascular nitric oxide production, promotes osteoclastogenesis, and contributes to the inflammatory microenvironment that sustains virtually every branch of the SGIC.

1.2 Gliadins and ATIs: The Permeability Drivers

Modern wheat contains two distinct classes of immunogenic proteins relevant to the SGIC framework. Gliadins — the alcohol-soluble fraction of wheat gluten — bind the chemokine receptor CXCR3 on intestinal epithelial cells and trigger the release of zonulin, which dismantles the tight junction proteins that seal the intestinal barrier. In individuals who consume wheat regularly, this produces a state of chronic intestinal hyperpermeability: the gate between the gut lumen and the systemic circulation is chronically ajar.

Amylase-trypsin inhibitors (ATIs) are non-gluten wheat proteins that act as potent activators of TLR4 on innate immune cells throughout the body. TLR4 activation drives NF- κ B-mediated production of TNF- α , IL-1 β , and IL-6 in gut-associated and systemic immune compartments, independently of celiac autoimmunity and independently of intestinal permeability. ATIs do not need to get through the gut wall to cause systemic inflammation — their immune activation signal distributes systemically through the lymphatic and portal circulation.

Together, gliadins and ATIs produce two convergent inflammatory inputs: a permeability-mediated amplification of Neu5Gc absorption and LPS translocation, and a direct innate immune activation that sustains systemic inflammatory tone. Wheat is not merely a gluten delivery vehicle. It is an immunological trigger in its own right, and its effects are operative in individuals who do not have celiac disease and do not test positive for gluten antibodies.

2. The Core Mechanism: The SGIC Pathway

The Sialic-Glycan Inflammatory Cascade begins with the simultaneous consumption of Neu5Gc and wheat-derived ATIs/gliadins — which describes virtually every meal in a typical Western dietary pattern. The cascade proceeds as follows:

Stage	Event	Key Mediators	Consequence
1	Dietary Neu5Gc absorbed; incorporated into cell surface glycoproteins throughout body	Anti-Neu5Gc IgG; complement activation; NF-κB	Xenosialitis: chronic low-grade systemic inflammation at all Neu5Gc-incorporation sites
2	Gliadin binds CXCR3; zonulin released; intestinal tight junctions disrupted	Zonulin; ZO-1, occludin, claudin-1 degradation	Intestinal hyperpermeability; enhanced Neu5Gc paracellular absorption; LPS translocation
3	ATIs activate TLR4 on innate immune cells throughout body including CNS microglia	TNF-α; IL-1β; IL-6; IL-8	Systemic and neuroinflammatory innate immune activation; NF-κB amplification
4	Systemic cytokine milieu (TNF-α, IL-6, IL-1β) distributed to all tissues	Endothelial NF-κB; osteoclast RANKL; NaPi-2a suppression; eNOS suppression; beta-cell IL-1β	Downstream branch activation: which branch is most clinically visible depends on individual susceptibility
5	Ammonium compounds co-present in processed foods lower mast cell degranulation threshold	NLRP3 inflammasome; IL-18; histamine	Potential of Stages 1–3; accelerated cascade timeline; more severe presentations
6	Concentrated bovine fats deliver Neu5Gc via chylomicron-mediated lymphatic route bypassing hepatic first-pass	Chylomicrons; MFGM glycoproteins; mesenteric lymphatics	Higher effective Neu5Gc antigenic load; deeper lymphatic sequestration; more severe and persistent presentations
7	Chronic olfactory-limbic positive valence toward triggering foods sustains dietary antigen intake	Neurobiological craving cycle	Biochemical trap: the foods driving the condition are actively sought by the compromised system

Table 1. The Core SGIC Pathway. Stages 1–4 are the universal mechanism active in all susceptible individuals consuming both antigens. Stages 5–6 amplify severity. Stage 7 explains why elimination is difficult and why understanding the mechanism matters.

3. The Downstream Branches: Conditions Within the Framework

The SGIC core mechanism produces different clinical presentations depending on which tissue or organ system is most susceptible in any given individual. The downstream branches are not separate diseases with separate mechanisms — they are different expressions of the same upstream inflammatory cascade in different biological territories. The white papers in this series address each branch in full mechanistic detail.

Condition	Primary SGIC Mechanism	Key Mediators	White Paper
Acne & Rosacea	Cutaneous TRM cell activation; lymphatic sequestration; histamine-driven sebaceous response; nocturnal amplification	Anti-Neu5Gc IgG; histamine; TRM cells; IL-6; TNF- α	Hamer Loop Manuscript Series (Documents 1–6); Cutaneous Immune Phenotype
Oily Skin (Seborrhoea)	Histamine-induced vasodilation; thermal sebaceous stimulation; mechanical interstitial displacement	Histamine; eNOS; interstitial pressure	Hamer Loop Document 5
Irritable Bowel / Gastrointestinal Sensitivity	ATI-driven TLR4 intestinal activation; gliadin-mediated permeability; TRM cell-mediated rapid GI response	TLR4; zonulin; GI TRM cells; TNF- α ; IL-17	Celiac, IBS & Gastrointestinal Sensitivity White Paper; Arthritis/Soup Flare (GI TRM section)
Celiac Disease / NCGS	HLA-DQ2/DQ8-mediated adaptive immune response to gliadin; ATI-driven innate activation; Neu5Gc-primed permeability amplifying gliadin translocation	HLA-DQ2/DQ8; tTG antibodies; ATI/TLR4; anti-Neu5Gc IgG	Celiac, IBS & Gastrointestinal Sensitivity White Paper; Cutaneous Immune Phenotype
Inflammatory Arthritis / Periarticular Inflammation	Immune complex synovial sequestration; gliadin molecular mimicry with type II collagen; complement-driven synovitis	Anti-Neu5Gc IgG; C3/C4; gliadin cross-reactive T cells; C5a	Arthritis/Soup Flare (Gastro-Arthritic Branch)
Osteoporosis	Osteoclastogenesis via TNF- α /IL-6/RANKL; calcium malabsorption via ATI-damaged epithelium; periosteal interstitial inflammatory microenvironment	RANKL; TNF- α ; IL-6; TRPV6; NaPi-2a; periosteal stem cells	Dietary Xenoantigens and Osteoporosis
Hypertension & Dyslipidaemia	Endothelial xenosialitis; eNOS suppression; NO depletion; LPS-driven TLR4 endothelial activation; atherogenic cytokine-driven lipid remodelling	Anti-Neu5Gc IgG; eNOS; VCAM-1; ICAM-1; TNF- α ; IL-6; CETP; apoA-I	Dual Dietary Antigen Exposure as Driver of Hypertension & Dyslipidaemia

Condition	Primary SGIC Mechanism	Key Mediators	White Paper
Obesity & Type 2 Diabetes	Gut dysbiosis; LPS metabolic endotoxaemia; leptin resistance via hypothalamic neuroinflammation; adiponectin suppression; beta-cell xenosialitis and apoptosis	Anti-Neu5Gc IgG; Firmicutes/Bacteroidetes ratio; SOCS3; adiponectin; IL-1 β ; PI3K-AKT	Neu5Gc as Primary Driver of Obesity & T2DM
Cancer	Xenosialitis-driven tumour angiogenesis (VEGF/Wnt); glycocalyx Neu5Gc shield enabling immune evasion; ATI-driven immune drain reducing anti-tumour surveillance	Anti-Neu5Gc IgG; VEGF; Wnt/ β -catenin; Siglec receptors; NK cell dysfunction	Dietary Elimination of Neu5Gc & ATIs as Cancer Surveillance Potentiator
Lyme Disease / PTLDS	Lymphatic functional overload impairing Borrelia clearance; microvascular endothelial dysfunction reducing immune access to sanctuary sites; neuroinflammatory brain fog amplification	Anti-Neu5Gc IgG; lymphatic stasis; eNOS; TLR4/LPS; microglial activation	Dietary Xenoantigen Load as Amplifier of Chronic Lyme Disease Sequelae
Fibromyalgia	ATI-driven microglial TLR4 activation; central sensitisation; inflammatory NaPi-2a suppression driving phosphate accumulation; xenosialitis musculotendinous interstitial load	TLR4; microglial TNF- α /IL-1 β ; NaPi-2a; NMDA sensitisation; interstitial pressure at tender points	Dietary Xenoantigens as Primary Drivers of Fibromyalgia
FSHD	Sarcolemmal xenosialitis driving oxidative stress promoting DUX4 reactivation; ATI/Neu5Gc TNF- α suppressing MyoD and satellite cell repair	Anti-Neu5Gc IgG; NADPH oxidase; ROS; D4Z4 methylation; TNF- α ; MyoD; PAX7	Dietary Xenoantigens as Modulating Variables in FSHD
Psoriasis	Gliadin-driven immune activation at keratinocyte surfaces; Neu5Gc xenosialitis amplifying IL-17/IL-23 psoriatic pathway; dietary pattern correlation with flare and remission	IL-17; IL-23; TNF- α ; gliadin cross-reactive T cells; anti-Neu5Gc IgG	Psoriasis, Neu5Gc & Gluten Case Report
Dermatitis Herpetiformis	Gliadin IgA-mediated immune complex deposition at dermal papillae; Neu5Gc xenosialitis amplifying cutaneous inflammatory burden	Gliadin IgA; tTG; anti-Neu5Gc IgG; complement at dermo-epidermal junction	Celiac, IBS & Gastrointestinal Sensitivity White Paper

Table 2. Downstream Branches of the SGIC Framework. Each condition shares the same upstream antigen sources and the same core NF- κ B inflammatory pathway. The branch that

becomes most clinically visible is determined by individual genetic susceptibility, immune phenotype, and dietary antigen exposure pattern.

4. The Two Immune Phenotypes

Not everyone who consumes a Western diet develops visible inflammatory disease. The Cutaneous Immune Phenotype paper establishes a framework for understanding this variability: two broad immune phenotypes are proposed in response to the combined dietary burden of Neu5Gc and gluten-derived antigens.

Phenotype A — the Reactive phenotype — is characterised by robust HLA-mediated immune reactivity that mounts a vigorous response to both dietary antigens. These individuals develop the visible cutaneous manifestations — acne, rosacea, dermatitis herpetiformis — that mark them as immunologically reactive. They are the canaries. Their visible symptoms are a signal of an immune system that is doing its job, loudly and at visible cost.

Phenotype B — the Tolerant phenotype — is characterised by chronically downregulated immune surveillance, a state of acquired immune tolerance driven by persistent xenosialitis. These individuals appear to tolerate the same dietary antigens without cutaneous manifestation. They are not healthier. They are quieter. The same inflammatory processes are operating in their tissues, but without the visible cutaneous signal that would prompt investigation. They may be accruing silent cardiovascular, metabolic, and neoplastic damage without any early warning system to alert them or their clinicians.

The most important clinical implication of this framework is counterintuitive: the absence of acne or rosacea does not indicate dietary antigen tolerance. It may indicate immune exhaustion. The patients who should most urgently be asked about their Neu5Gc and ATI burden may be the ones whose immune systems have become too quiet to raise the alarm.

5. The Severity Determinants

Within the SGIC framework, the severity of any individual's inflammatory presentation is determined by three interacting variables, each addressed in detail in the Hamer Loop Manuscript Series:

5.1 Antigen Load: What and How Much

Not all sources of Neu5Gc produce equivalent inflammatory responses. The severity hierarchy established in the Neu5Gc Severity Index (Appendix II of the Hamer Loop Manuscript Series) reflects the combined effects of antigen concentration and delivery route. Concentrated bovine fats — delivered via chylomicron-mediated lymphatic entry that bypasses hepatic first-pass metabolism — produce more severe and persistent presentations than whole-food dairy

sources. Processed foods containing both Neu5Gc sources and ammonium catalysts produce the most severe presentations of all.

5.2 Amplification: The Ammonium Potentiation Effect

Ammonium-containing compounds — caramel colour Classes III and IV, natural and artificial flavour systems, enriched flour, and processed flavouring agents — lower the mast cell degranulation threshold and prime the NLRP3 inflammasome, producing inflammatory responses disproportionate to the Neu5Gc load alone. The processed food supply is saturated with these compounds, making the majority of commercially produced foods in Western markets simultaneously Neu5Gc sources, ATI/gliadin sources, and ammonium potentiators — triple-threat trigger profiles that produce the most severe and rapidly developing SGIC presentations.

5.3 Delivery Route: The Fat Amplifier

Dietary fat determines whether Neu5Gc enters the systemic circulation via the portal route — subject to hepatic first-pass attenuation — or via the chylomicron-mediated lymphatic route that deposits antigen directly into the lymphatic system, bypassing the liver entirely. High-fat bovine sources deposit their Neu5Gc load into exactly the compromised lymphatic clearance system that the Hamer Loop proposes as the site of inflammatory sequestration. This is why the severity hierarchy correlates with fat concentration rather than Neu5Gc content alone: the delivery route matters as much as the antigen quantity.

6. The Practical Tools: Acnerase and the Severity Index

The Hamer Glycan Systems framework is not solely theoretical. Its translational output is the Acnerase food-label scanning application and the Neu5Gc Severity Index (Appendix II of the Hamer Loop Manuscript Series).

Acnerase is a free Android application that uses a proprietary database of 150+ ingredient synonyms to identify Neu5Gc sources, ammonium catalysts, and ATI/gliadin indicators in commercial food products through real-time label scanning. It does not require the user to understand the mechanism — it identifies the trigger ingredients before consumption, enabling informed avoidance regardless of whether the user is managing acne, rosacea, arthritis, fibromyalgia, or any other SGIC branch condition.

The Neu5Gc Severity Index classifies 148 specific ingredients across a five-tier severity rating with resolution categories, providing the mechanistic backbone for Acnerase's risk stratification output. It is documented in full in Document 7 / Appendix II of the Hamer Loop Manuscript Series.

The Live Demonstration Protocol (Document 6 of the Hamer Loop Manuscript Series) provides a structured, four-phase observational framework for witnessing the SGIC in real time: baseline establishment, predictive induction with anatomical pre-specification of lesion sites, observed cascade progression, and targeted resolution. It is the bridge between theoretical framework and empirical observation that the framework's evidential position requires.

7. The Document Architecture

The full Hamer Glycan Systems research corpus comprises the following documents, all available through hamerglycans.com:

Document	Type	Primary Content
Hamer Loop Document 1: The Hamer Loop & Xeno-Antigenic Sequestration	Foundation manuscript	Core mechanism; clinical timeline; TRM cells; systemic resolution profile; validation framework
Hamer Loop Document 2: Ammonium & Lesion Phase Classification	Mechanism detail	Ammonium Potentiation Effect; three-phase lesion classification; severity hierarchy
Hamer Loop Document 3: Dietary Fats & Neu5Gc Delivery	Mechanism detail	Chylomicron-mediated lymphatic delivery; MFGM; negative controls (lard, salmon)
Hamer Loop Document 4: Whole-Food Sources	Mechanism detail	Baseline tier; portal absorption; absence-of-amplifier principle; re-exposure sensitivity
Hamer Loop Document 5: Oily Skin as Physiological Response	Mechanism detail	Seborrhoea as downstream consequence; thermal mechanism; mechanical displacement; causal inversion
Hamer Loop Document 6: Live Demonstration Protocol	Validation protocol	Four-phase prospective self-provocation; olfactory inversion; gelatin-antihistamine paradox; biochemical sampling integration
Hamer Loop Document 7 / Appendix II: Neu5Gc Severity Index	Reference database	148 ingredients; severity 1–5; resolution categories A–E; Acnerase backbone
Cutaneous Immune Phenotype	Journal hypothesis paper	Two immune phenotypes; HLA convergence; canary framework; 6 testable predictions
SGIC Master Overview	This document	Unified framework; all branches; severity determinants; document architecture
Celiac, IBS & Gastrointestinal Sensitivity	Condition white paper	GI branch; celiac spectrum; dermatitis herpetiformis; gut TRM cells; elimination protocol
Dietary Xenoantigens and Osteoporosis	Condition white paper	Triple-pathway model; calcium malabsorption; periosteal microenvironment; Subject B (Goldie) case

Document	Type	Primary Content
Dietary Xenoantigens as Modulating Variables in FSHD	Condition white paper	DUX4 oxidative gate; sarcolemmal xenosialitis; satellite cell suppression; 12-week self-experiment
Dietary Xenoantigen Load as Amplifier of Chronic Lyme Disease Sequelae	Condition white paper	Persistent Fire model; lymphatic overload; microvascular dysfunction; brain fog mechanism
Dietary Xenoantigens as Primary Drivers of Fibromyalgia	Condition white paper	ATI/microglial sensitisation; NaPi-2a phosphate transport; guaifenesin protocol context; Stephanie case; ASD dimension
Dual Dietary Antigen Exposure as Driver of Hypertension & Dyslipidaemia	Condition white paper	Endothelial xenosialitis; eNOS impairment; atherogenic lipid phenotype; pharmacokinetic interaction
Neu5Gc as Primary Driver of Obesity & T2DM	Condition white paper	Four-pathway metabolic cascade; leptin resistance; beta-cell toxicity; dairy elimination paradox
Dietary Elimination of Neu5Gc & ATIs as Cancer Surveillance Potentiator	Condition white paper	Glycocalyx shield; xenosialitis tumour growth; immune reconstitution; Neu5Ac substitution; biotherapeutic synergy
Hamer Loop — Gastro-Arthritic Branch (Soup Flare)	Case study	Triple-threat sentinel observation; synovial sequestration; GI TRM cells; liquid delivery accelerant
Psoriasis, Neu5Gc & Gluten Case Report	Case report	Dietary correlation with flare/remission; IL-17/IL-23 pathway; Neu5Gc amplification of psoriatic cascade

Table 3. Hamer Glycan Systems Document Architecture. All documents available at hamerglycans.com. The Hamer Loop Manuscript Series (Documents 1–7) provides the mechanistic foundation. The condition white papers apply the framework to specific clinical presentations. The SGIC Master Overview (this document) maps the full landscape.

8. The Six-Investigation Validation Framework

The complete validation framework for the SGIC hypothesis is documented in Section 10 of Hamer Loop Document 1. It comprises six investigations, each targeting a specific mechanistic claim and each falsifiable. They are summarised here for reference:

Investigation	Target Claim	Method
10.1 Serum mast cell tryptase during active flare	Systemic mast cell degranulation as Stage 1 mechanism	Venepuncture at T+40 minutes during controlled provocation; tryptase vs baseline
10.2 Complement fractions C3/C4	Immune complex-mediated complement consumption as initiating event	Serial draws at T+0, T+2hrs, T+4–6hrs; C3/C4 depression during flare vs baseline
10.3 Anti-Neu5Gc antibody titres	Titre correlation with flare severity and dietary exposure	Validated ELISA; titre elevation on provocation; progressive decline on elimination
10.4 Skin biopsy at 2–4 hours	IgG/C3 immune complex deposition in early lesion tissue	Direct immunofluorescence staining; early-phase biopsy vs established lesion comparison
10.5 Urinary Neu5Gc quantification	Differential Neu5Gc absorption in susceptible vs non-susceptible individuals	Non-invasive; before/during/after controlled provocation; correlation with flare onset
10.6 Near-infrared fluorescence lymphatic imaging during active flare	Regional lymphatic stasis as sequestration mechanism	NIRF imaging during flare vs cleared baseline; lymphatic transit velocity comparison

Table 4. The Six-Investigation Validation Framework. Each investigation is independently executable with existing clinical and laboratory tools. Together they constitute a comprehensive evidentiary programme for the SGIC core mechanism.

9. The Single Intervention

The SGIC framework, across all its branches and all its conditions, converges on a single practical intervention: the identification and elimination of dietary Neu5Gc sources and wheat-derived ATIs and gliadins from the diet.

This intervention is: free; available to anyone without a prescription, a clinical referral, or a research protocol; low-risk in the sense that eliminating bovine red meat, bovine dairy, and wheat from the diet has no known adverse health consequences when nutritional needs are met through alternative sources; and potentially consequential across multiple organ systems simultaneously, because it addresses the upstream source of the entire cascade rather than one downstream consequence of it.

It is also difficult. The foods that must be eliminated are among the most heavily marketed, most socially embedded, most institutionally supported, and most neurobiologically craved foods in the Western dietary pattern. The craving for these foods is not a simple matter of preference — it is, within this framework, a biochemical consequence of chronic antigen exposure that generates positive olfactory-limbic valence toward the triggering foods. Understanding this

mechanism does not make elimination easy. But it makes it comprehensible. And comprehension is the first step toward change that the patient — rather than the clinician — chooses to make.

The Acnerase application exists to make the practical side of that choice easier. Hidden sources of both antigens are pervasive in the commercial food supply, listed under label terms that do not announce their immunological relevance. The Acnerase ingredient database exists to surface those hidden sources, in real time, before the food is consumed. It does not make dietary decisions for the user. It makes informed decisions possible.

10. What the Canary Knew

The skin is the most visible branch of the SGIC. For the 1.2 billion people living with acne or rosacea, it is the branch that drives them to dermatologists, to pharmacies, to the mirror, and in some cases to profound distress about their appearance and their health. It is also, within this framework, the branch that carries the most information — because a skin that is inflaming in response to specific dietary inputs is a skin that is telling you something about what is happening inside the body at a systemic level.

The canary in the coal mine was not the problem. The canary was the signal. The miners who paid attention to the canary survived. The ones who treated the canary's distress as an inconvenience to be managed — or as a cosmetic problem to be concealed — did not benefit from the information the canary was trying to provide.

The acne is the canary. The rosacea is the canary. The fibromyalgia tender point is the canary. The arthritic morning stiffness is the canary. The blood pressure that will not fully respond to medication is the canary. The bone density scan that keeps declining despite the bisphosphonate is the canary. The fatigue that twenty years of management has not resolved is the canary.

All of them are saying the same thing. They are saying: something in the environment of this body is producing chronic inflammation, and the inflammation is expressing itself in the tissues most vulnerable to it, most susceptible to it, most mechanically or genetically predisposed to carry its burden.

The mine is the body. The coal is the Western dietary pattern. And the two compounds that are doing the most damage — the Neu5Gc from the bovine dairy and red meat, and the ATIs and gliadins from the wheat — are in virtually every meal, quietly sustaining every branch of the cascade, while the canary keeps singing and the clinician keeps treating the bird.

This framework is an invitation to listen to the canary.

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